

Learnability

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following Shalev-Shwartz & Ben-David, *Understanding Machine Learning* (2014), Part I

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Abstract

When is a learning problem solvable from finitely many examples, and what does the answer depend on? This note develops the answer end to end: the formal model and ERM, why unrestricted learning overfits, PAC learnability with a complete proof for finite classes, uniform convergence as the engine behind agnostic learning, the No-Free-Lunch theorem (there is no universal learner), and the VC dimension, which characterizes learnability exactly. Companion experiments: `learnability_demo.py` exhibits the memorizer overfitting, the impossibility of shattering two points with thresholds, and the two learning rates $1/m$ and $1/\sqrt{m}$ of the Fundamental Theorem.

1 The formal model

Definition 1 (The learning problem). A domain $\mathcal{X}$ (instances), a label set $\mathcal{Y} = \{0, 1\}$, and an unknown distribution $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$. The learner receives a training sample $S = ((x_1, y_1), \dots, (x_m, y_m))$ drawn i.i.d. from $\mathcal{D}$, and must output a hypothesis $h : \mathcal{X} \rightarrow \mathcal{Y}$. Its quality is the true risk (generalization error) and its observable proxy, the empirical risk:

$$L_{\mathcal{D}}(h) = \Pr_{(x,y) \sim \mathcal{D}}[h(x) \neq y], \quad L_S(h) = \frac{1}{m} |\{i : h(x_i) \neq y_i\}|. \quad (1)$$

The learner never sees $\mathcal{D}$; everything it knows is S . In the *realizable* setting one additionally assumes a true labeling function f with $y = f(x)$ and some hypothesis achieving zero risk; the general (*agnostic*) setting assumes nothing.

Definition 2 (Empirical Risk Minimization). Given a hypothesis class $\mathcal{H} \subseteq \mathcal{Y}^{\mathcal{X}}$ chosen before seeing data, the ERM rule outputs

$$h_S \in \arg \min_{h \in \mathcal{H}} L_S(h). \quad (2)$$

2 Why restrict $\mathcal{H}$ at all: overfitting

Let ERM range over *all* functions. Consider the memorizer

$$h_S(x) = \begin{cases} y_i & \text{if } x = x_i \text{ for some } i, \\ 0 & \text{otherwise.} \end{cases}$$

Its empirical risk is 0 by construction. But if $\mathcal{X}$ is continuous and $\mathcal{D}$ has a density, a fresh x almost surely differs from every training point, so h_S predicts the constant 0 on all new data — with, e.g., $L_{\mathcal{D}}(h_S) = \frac{1}{2}$ when labels are balanced. Perfect memory, zero learning (first panel of the demo). The conclusion is structural, not a technicality: *some prior restriction — an inductive bias — is necessary*. Sections 5–6 make this precise (No-Free-Lunch) and quantify exactly how much restriction is needed (VC dimension).

3 PAC learnability, and the finite realizable case

Definition 3 (PAC learnability, realizable). $\mathcal{H}$ is PAC learnable if there exist a function $m_{\mathcal{H}}(\varepsilon, \delta)$ and a learner such that for every $\varepsilon, \delta \in (0, 1)$ and every realizable $\mathcal{D}$: given $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$ i.i.d. examples, the output h_S satisfies

$$\Pr[L_{\mathcal{D}}(h_S) \leq \varepsilon] \geq 1 - \delta. \quad (3)$$

Probably $(1 - \delta)$, over the random draw of S) Approximately (ε) Correct. Both slacks are unavoidable: a finite sample can be unlucky (δ) and can never pin $\mathcal{D}$ down exactly (ε) .

Theorem 1 (Finite classes are PAC learnable). Every finite $\mathcal{H}$ is PAC learnable by ERM with sample complexity

$$m_{\mathcal{H}}(\varepsilon, \delta) \leq \left\lceil \frac{\log(|\mathcal{H}|/\delta)}{\varepsilon} \right\rceil. \quad (4)$$

Proof. Call h bad if $L_{\mathcal{D}}(h) > \varepsilon$, and let $\mathcal{H}_B \subseteq \mathcal{H}$ be the bad hypotheses. A fixed bad h labels one random example correctly with probability at most $1 - \varepsilon$, so it *survives* the sample (i.e. $L_S(h) = 0$) with probability

$$\Pr[L_S(h) = 0] \leq (1 - \varepsilon)^m \leq e^{-\varepsilon m}.$$

By the union bound, the probability that *some* bad hypothesis survives is at most $|\mathcal{H}| e^{-\varepsilon m}$, which is $\leq \delta$ once $m \geq \log(|\mathcal{H}|/\delta)/\varepsilon$. On the complementary event no bad hypothesis is consistent with S ; and by realizability the perfect h^* is consistent, so ERM (2) returns a consistent hypothesis — necessarily a good one: $L_{\mathcal{D}}(h_S) \leq \varepsilon$. $\square$

Read the bound: examples are priced logarithmically in the class size and linearly in $1/\varepsilon$ — doubling the class costs one more $\varepsilon^{-1} \log 2$ worth of data, not double the data.

4 Agnostic learning via uniform convergence

Realizability is a strong fiction; drop it. The target now is *competing with the best in the class*.

Definition 4 (Agnostic PAC). $\mathcal{H}$ is agnostically PAC learnable if with $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$ samples, for every $\mathcal{D}$,

$$\Pr\left[L_{\mathcal{D}}(h_S) \leq \min_{h \in \mathcal{H}} L_{\mathcal{D}}(h) + \varepsilon\right] \geq 1 - \delta. \quad (5)$$

The engine is a property of the *sample*, not the algorithm:

Definition 5 (ε -representative sample; uniform convergence). S is ε -representative for $\mathcal{H}$ if $|L_S(h) - L_{\mathcal{D}}(h)| \leq \varepsilon$ for all $h \in \mathcal{H}$ simultaneously. $\mathcal{H}$ has the uniform convergence property if some $m_{\mathcal{H}}^{\text{UC}}(\varepsilon, \delta)$ samples make S ε -representative with probability $\geq 1 - \delta$, for every $\mathcal{D}$.

Lemma 1 (UC $\Rightarrow$ agnostic PAC via ERM). *If S is $\frac{\varepsilon}{2}$ -representative, then the ERM output satisfies $L_{\mathcal{D}}(h_S) \leq \min_{h \in \mathcal{H}} L_{\mathcal{D}}(h) + \varepsilon$.*

Proof. For any $h \in \mathcal{H}$: $L_{\mathcal{D}}(h_S) \leq L_S(h_S) + \frac{\varepsilon}{2} \leq L_S(h) + \frac{\varepsilon}{2} \leq L_{\mathcal{D}}(h) + \varepsilon$, using representativeness, then ERM's optimality on S , then representativeness again. $\square$

Uniform convergence is precisely what licenses trusting the training score of a hypothesis *selected because of* its training score — the trap the memorizer fell into.

Theorem 2 (Finite classes are agnostically learnable). *Every finite $\mathcal{H}$ has uniform convergence, hence is agnostically PAC learnable by ERM, with*

$$m_{\mathcal{H}}(\varepsilon, \delta) \leq \left\lceil \frac{2 \log(2|\mathcal{H}|/\delta)}{\varepsilon^2} \right\rceil. \quad (6)$$

Proof. Fix h . The random variables $\mathbf{1}[h(x_i) \neq y_i]$ are i.i.d. in $[0, 1]$ with mean $L_{\mathcal{D}}(h)$, so Hoeffding's inequality gives $\Pr[|L_S(h) - L_{\mathcal{D}}(h)| > \varepsilon] \leq 2e^{-2m\varepsilon^2}$. A union bound over $\mathcal{H}$ makes the failure probability $\leq 2|\mathcal{H}|e^{-2m\varepsilon^2} \leq \delta$ for m as in (6) (at accuracy $\varepsilon/2$, whence the constant 2). Conclude with Lemma 1. $\square$

Note the price of dropping realizability: $1/\varepsilon$ became $1/\varepsilon^2$ — estimating a mean to accuracy ε costs ε^{-2} , whereas certifying perfection costs only ε^{-1} . These are the $1/m$ vs $1/\sqrt{m}$ rates the demo measures.

5 No Free Lunch: bias is necessary

Theorem 3 (No-Free-Lunch). *Let A be any learner for binary classification over a domain $\mathcal{X}$, and let $m \leq |\mathcal{X}|/2$. Then there exists a distribution $\mathcal{D}$, realizable by some $f : \mathcal{X} \rightarrow \{0, 1\}$, on which*

$$\Pr_{S \sim \mathcal{D}^m} \left[L_{\mathcal{D}}(A(S)) \geq \frac{1}{8} \right] \geq \frac{1}{7}. \quad (7)$$

(Proof idea: put uniform mass on $2m$ points; among all labelings of those points, the sample reveals at most half, and on the unseen half any fixed prediction is wrong for half the labelings in expectation — so *some* labeling defeats A .)

Corollary 1. *If $\mathcal{X}$ is infinite, the class of all functions $\mathcal{Y}^{\mathcal{X}}$ is not PAC learnable.*

There is no universal learner: an algorithm succeeding on every problem with a fixed budget does not exist, because for any budget an adversarial (yet realizable!) task exists. Learning therefore always starts from prior knowledge — the choice of $\mathcal{H}$. The cost structure is the *bias-complexity decomposition*: for the ERM output,

$$L_{\mathcal{D}}(h_S) = \underbrace{\min_{h \in \mathcal{H}} L_{\mathcal{D}}(h)}_{\text{approximation error}} + \underbrace{L_{\mathcal{D}}(h_S) - \min_{h \in \mathcal{H}} L_{\mathcal{D}}(h)}_{\text{estimation error}}, \quad (8)$$

richer classes lowering the first term and (by (6) and its VC analogue below) raising the second. Overfitting and underfitting are the two ways of losing this trade.

6 The VC dimension characterizes learnability

Finiteness of $\mathcal{H}$ was sufficient but is far from necessary — thresholds on $\mathbb{R}$ form an infinite class that is obviously learnable. The right measure counts *behaviors*, not hypotheses.

Definition 6 (Shattering; VC dimension). For $C = \{c_1, \dots, c_k\} \subseteq \mathcal{X}$, the restriction of $\mathcal{H}$ to C is $\mathcal{H}_C = \{(h(c_1), \dots, h(c_k)) : h \in \mathcal{H}\} \subseteq \{0, 1\}^k$. C is *shattered* if $\mathcal{H}_C = \{0, 1\}^k$ — every labeling of C is realized by some hypothesis. The VC dimension $\text{VCdim}(\mathcal{H})$ is the size of the largest shattered set (infinite if arbitrarily large sets are shattered).

Example 1.

- **Thresholds** $\{x \mapsto \mathbf{1}[x > a]\}$ on $\mathbb{R}$: any single point is shattered; for $x_1 < x_2$ the labeling $(1, 0)$ is impossible (the demo checks this exhaustively). $\text{VCdim} = 1$.
- **Intervals** $\mathbf{1}[a \leq x \leq b]$: two points shatter; three fail on the labeling $(1, 0, 1)$. $\text{VCdim} = 2$.
- **Axis-aligned rectangles** in $\mathbb{R}^2$: $\text{VCdim} = 4$.
- **Halfspaces** in $\mathbb{R}^d$: $\text{VCdim} = d + 1$.
- **Finite classes**: $\text{VCdim}(\mathcal{H}) \leq \log_2 |\mathcal{H}|$ (shattering k points needs 2^k hypotheses).
- **One parameter, infinite VC**: $\{x \mapsto \lceil \sin(\theta x) \rceil : \theta \in \mathbb{R}\}$ shatters arbitrarily large sets — capacity is not parameter count.

Why finite VC tames infinite classes: the *growth function* $\tau_{\mathcal{H}}(m) = \max_{|C|=m} |\mathcal{H}_C|$ counts distinguishable behaviors on m points, and once m exceeds the VC dimension it stops growing exponentially:

Lemma 2 (Sauer-Shelah). If $\text{VCdim}(\mathcal{H}) = d < \infty$, then $\tau_{\mathcal{H}}(m) \leq \sum_{i=0}^d \binom{m}{i}$, and for $m \geq d$,

$$\tau_{\mathcal{H}}(m) \leq \left(\frac{em}{d}\right)^d \quad \text{— polynomial, not } 2^m. \quad (9)$$

An effective “size” of $(em/d)^d$ plugged into finite-class arguments (with a symmetrization step) yields uniform convergence for any finite-VC class — and NFL supplies the converse: shattered sets of unbounded size embed the unlearnable “all functions” problem.

Theorem 4 (Fundamental Theorem of Statistical Learning). For a class $\mathcal{H}$ of binary classifiers, the following are equivalent:

1. $\mathcal{H}$ has the uniform convergence property;
2. ERM is a successful agnostic PAC learner for $\mathcal{H}$;
3. $\mathcal{H}$ is agnostically PAC learnable;
4. $\mathcal{H}$ is PAC learnable (realizable);
5. $\text{VCdim}(\mathcal{H}) = d < \infty$.

Quantitatively, with constants absorbed in Θ :

$$\boxed{m_{\mathcal{H}}^{\text{realizable}}(\varepsilon, \delta) = \Theta\left(\frac{d + \log(1/\delta)}{\varepsilon}\right), \quad m_{\mathcal{H}}^{\text{agnostic}}(\varepsilon, \delta) = \Theta\left(\frac{d + \log(1/\delta)}{\varepsilon^2}\right).} \quad (10)$$

This is the punchline of the whole subject: *learnability is a combinatorial property of the hypothesis class* — one number, the VC dimension, decides yes or no and prices the data; the algorithm (ERM) is universal among learnable classes. The demo’s log-log panel shows the rates live for thresholds ($d = 1$): the realizable excess risk decays like $1/m$, and under label noise the uniform-convergence supremum $\sup_{h \in \mathcal{H}} |L_S(h) - L_{\mathcal{D}}(h)|$ — representativeness over the whole class, the quantity that costs ε^{-2} — decays like $1/\sqrt{m}$.

Remark 1 (What learnability does not promise). (i) Computation: *the theorem counts samples, not time* — ERM can be NP-hard even for learnable classes (e.g. agnostic halfspaces); statistical and computational learnability are separate axes. (ii) Distribution-specific help: (10) is a worst-case-over- $\mathcal{D}$ statement, and the demo catches this in the act: with uniform 10% label noise the measured excess risk of ERM decays like $1/m$, faster than the worst-case $1/\sqrt{m}$, even though the uniform-convergence supremum genuinely decays like $1/\sqrt{m}$ on the same data. The reason: uniform noise keeps the noise level away from $\frac{1}{2}$ (a margin condition), and the linear risk penalty stops ERM from wandering into the region where the empirical fluctuations that drive the supremum live. Favorable distributions, margins, and regularization all beat the worst case; that is where the rest of learning theory (Tsybakov noise conditions, nonuniform learnability, Rademacher complexity, stability) begins.